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Review Article

Developments and Trends of Molecularly Imprinted Solid-Phase Microextraction

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This review focuses on a solid-phase microextraction (SPME) method coupled with molecularly imprinted polymers (MIPs), namely molecularly imprinted solid-phase microextraction (MISPME). The first two sections discuss the summaries of conventional SPME and MIPs. The third section reviews the development of MISPME in past years, including the preparation of MISPME, and the applications to compounds in real samples.

Introduction

The determination of organic contaminants presenting at trace levels in complex matrices requires a procedure of sample pretreatment. Historically, liquid–liquid extraction (LLE) (1) and solid-phase extraction (SPE) (2) were often used for the extraction of analytes from aqueous matrices. However, LLE is time-consuming, generally labor-intensive and requires large quantities of expensive, toxic and environmentally unfriendly organic solvents. SPE needs less solvent but still requires multiple steps. Additionally, the enrichment efficiency of SPE is relatively low. Therefore, the simplification and miniaturization of sample preparation methods are recommended, which have the advantages of using either no or very little amounts of toxic organic solvents. Since its introduction by Arthur and Pawliszyn in 1990 (3), solid-phase microextraction (SPME) has developed rapidly and become widely used for the analysis of food (4–6), biological (7, 8), environmental (9, 10) and pharmaceutical samples (11, 12). This is because this simple and solvent-free sample preparation technique, which combines extraction, concentration and sample introduction in one step, is portable, sensitive and convenient to couple with various analytical instruments, such as gas chromatography (GC).

SPME is based on the partitioning of analytes between the sample and the coating, which is fixed on the surface of a fused-silica fiber or metal wire. Therefore, an excellent material for coating is very important to obtain high extraction efficiency. Until now, several SPME coating materials have been commercially available, including polydimethylsiloxane (PDMS), carboxen/PDMS, PDMS/divinylbenzene (DVB), polyacrylate (PA), carbowax/DVB and carbowax/templated resin (13). In addition, several laboratory-made SPME fibers prepared by new approaches have been reported, including sol-gel technology, physical deposition, electrochemical procedure and vapor deposition technology. These coated fibers, which possess high extraction capacity and provide good analytical precision, have been successfully applied in some cases of real sample analysis. Despite these cited advantages, the primary drawback associated with these is the lack of selectivity of the coatings during the extraction of target analytes. Due to their unsatisfactory selectivity, these traditional coatings usually cannot efficiently extract analytes in complex biological or environmental samples.

In recent years, molecularly imprinted polymers (MIPs) have attracted much attention due to their outstanding advantages, such as predetermined recognition ability, stability, relative ease and low cost of preparation and potential application to a wide range of target molecules (14). The selectivity of MIPs has been applied in several applications, such as sensors (15, 16), capillary electrochromatography (17, 18), enantiomeric separations (19, 20), SPE (21–23) and catalysis (24, 25). In the last few years, one of the most promising technical applications has been molecularly imprinted solid-phase microextraction (MISPME), which is a very advanced methodology that solves the problem of selectivity of the coatings (26, 27). Thus far, the technology of MISPME has not been reviewed. The present review reports the summaries of SPME and MIPs and describes the development of MISPME in recent years, including the preparation of MISPME and the applications to compounds in real samples.

SPME

Principle of SPME

In SPME, coated fibers are used to isolate and concentrate analytes into a range of coating materials. After extraction, the fibers are transferred, with the help of the syringe-like handling device, to analytical instruments for the separation and quantification of the target analytes. The procedure for collecting the sample is shown in Figure 1.

SPME is based on a partition mechanism and the establishment of equilibrium between the analyte and the sample matrix. The distribution constant, K, for equilibrium (10) is given by

$$K = \frac{C_f}{C_s} \quad (1)$$

where C_f and C_s denote equilibrium concentrations of the analyte in the fiber and the sample, respectively. Equation 1 means that the extracted amount is constant within the limits of experimental error after the establishment of equilibrium.

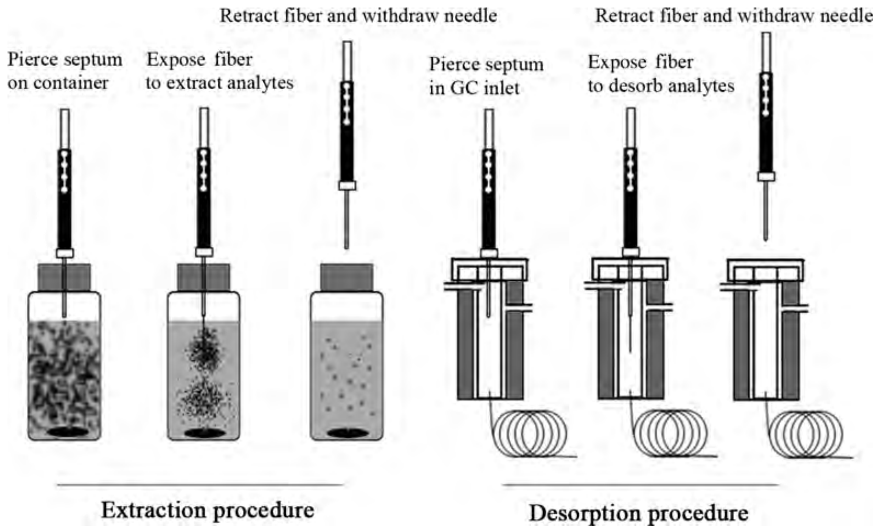

Figure 1. Procedure for collecting the sample using SPME.

In cases with a finite volume of the sample, Equation 1 can be expressed as

$$n = \frac{KV_f V_s C_o}{KV_f + V_s} \tag{2}$$

where n is the amount of the analyte extracted, C_o is the initial concentration of given analyte in the sample, V_f and V_s are the volume of fiber coating and the sample volume, respectively. Equation 2 indicates a proportional relationship between analyte concentration in the sample and the amount extracted by the coated fiber, which is the basis for analyte quantification. In cases with very large distribution constant, i.e., $K \gg V_s$, the equation can be written as

$$n = V_s C_o \tag{3}$$

Equation 3 indicates that the amount of adsorbed analyte is directly related to the volume of the sample and the initial concentration of the analyte with finite sample volume. In the field sampling, sample volume is very large and $V_s \gg KV_f$. Therefore, n can be written as

$$n = KV_f C_o \tag{4}$$

Equation 4 indicates that the amount extracted on fiber is directly related to volume of fiber coating and the concentration of the analyte, and is independent of the volume of the sample. Thus, the amount of extracted analyte will correspond directly to its concentration in the matrix.

Preparation of SPME coating

Since its introduction in the early 1990s, SPME has been extensively studied and the techniques for obtaining the SPME fiber coatings have been generated.

Sol-gel technology

Sol-gel technology has been an excellent SPME fiber coating-preparation method because of its inherent advantages and performance. These include a single-step manufacturing process, material homogeneity at the molecular level, chemical bonding between the sorbent and the fused-silica surface, high thermal and solvent stability and the porous structure of the hybrid material. Sol-gel-based SPME fibers were first introduced by Chong *et al.* (28) to improve the performance of traditional fibers. They used methyltrime-thoxysilane (MTMOS) as a precursor for preparing the chemically bonded PDMS fiber. Mackenzie (29) explained that the sol-gel network originating from an alkyl derivative of tetraalkoxysilane (TAOS) precursor possesses a more open structure and can effectively minimize the stress during drying and cracking. Since then, sol-gel technology used to prepare SPME fibers became a fast developing field. Subsequently, the preparation of unbreakable sol-gel fibers has been the focus of many researchers (30, 31). In 2006, the first unbreakable SPME fiber fabricated by sol-gel deposition was reported by Azenha and co-workers (32). They used titanium wire as a new, unbreakable substrate and showed that alkoxysilanes and hydrosilanes can form Ti-O-Si bonds under suitable conditions. In recent years, several attempts were made to obtain fibers with different characteristics using different coating polymers and modifiers (33–35).

Electrochemical procedure

Electrochemical procedures are an alternative way to deposit coatings on SPME fibers: the idea is to coat the metal fiber used in the SPME instrument with a conducting polymer, and cyclic voltammetry (CV) or potentiometry are usually used for this purpose. These techniques are inexpensive, and fiber coatings of the oxidized or reduced form of the polymer can also be obtained. The electrochemical procedures were first attempted by Wu *et al.* (36). They prepared polypyrrole (PPy) and poly-*N*-phenylpyrrole (PPPY) films on platinum as SPME fiber coatings. Since then, PPy and its derivatives have attracted great interest in the development of these procedures (37–39). Now, these electrochemically controlled SPME fibers can be used for preconcentration and matrix separation of anionic (40), cationic (41) and neutral analytes (42).

Physical deposition

One of the quickest and simplest means of coating an SPME fiber is to immerse it in a polymer solution, and then to condition or heat the fiber to stabilize it. This technology is called physical deposition. Farajzadeh and Rahmani (43) constructed an alumina-based SPME fiber using alumina powder and poly(vinyl chloride) (PVC) at the optimum ratio 97:3. Subsequently, Wang *et al.* (44) first reported multi-walled carbon nanotubes (MWCNTs) as SPME fiber coatings by using the approach of dispersing the MWCNTs in dimethylformamide (DMF) as suspension, and then dipping the fiber into the suspension, followed by heating at 160°C to remove the solvent. Recently, the authors' group (45) prepared a novel graphene (G)-based SPME fiber by immobilizing the synthesized G on stainless steel wire as a coating. The fiber was successfully applied to the determination of pyrethroids.

Molecularly Imprinted Techniques

Principle of molecularly imprinted techniques

MIPs are polymeric materials that exhibit high binding capacity and selectivity against a target molecule that is purposely present during the synthesis process. In the common approach, the synthesis of MIPs first involves the solution complexation of a template molecule with functional monomers, followed by polymerization of these monomers around the template with the aid of a cross-linker in the presence of an initiator. After the removal of the template by chemical reaction or extraction, binding sites are exposed that are complementary to the template in size, shape and position of the functional groups, which consequently allow its selective uptake. Figure 2 presents a scheme of the synthesis of MIPs.

Preparation of MIPs

Choice of polymerized materials

The primary polymerized materials involved in the production of an MIP are the template molecule, the functional monomer and the cross-linking agent.

In all molecular imprinting processes, the template molecule is of central importance because it directs the organization of the functional groups pendent to the functional monomers. The structure and functionalities of this molecule define the subsequent properties of the binding sites. Criteria to be considered when selecting a candidate molecule include its cost, its availability and the chemical functionalities defining its ability to interact strongly with monomers.

To obtain the highest affinity of the MIPs toward the target analyte, careful selection of the functional monomers used in the production of the MIP is crucial. Although there are many different commercially available functional monomers to choose from, the most widely used have been methacrylic acid (MAA) and 4-vinylpyridine (4-VP).

The third part involved in the synthesis of a MIP is the cross-linking agent, whose function is to deliver mechanical stability, stabilize the molecular recognition site and control the porosity of the polymer. Many cross-linkers compatible with molecular imprinting are known, many of which are commercially available and a few of which are capable of simultaneously complexing with the template, thus acting as functional monomers. The most widely used are ethylene glycol dimethacrylate (EDMA) and trimethylolpropane trimethacrylate (TRIM).

Polymerization techniques

Two different approaches to preparing MIPs have been reported: covalent and non-covalent approaches. Of the two approaches, the non-covalent approach has been used more extensively.

The covalent approach involves the formation of reversible covalent bonds between the template and monomers prior to polymerization. After the removal of the template by chemical reaction, the MIPs rebind template molecules via the same covalent interactions. The high stability of the template–monomer interaction leads to a homogenous population of binding sites, minimizing the existence of nonspecific sites (46). However, the difficulty of designing an appropriate template–monomer complex in which covalent bond formation and cleavage are readily reversible under mild conditions makes this approach rather restrictive.

The most widely used technique for preparing MIPs is the non-covalent approach, which was introduced by Arshady and Mosbach (47). This approach is based on the formation of relatively weak non-covalent interactions, such as hydrogen bonding, electrostatic forces, van der Waals forces or hydrophobic interactions between the template molecule and selected monomers before polymerization. There are several advantages of this technique, including easy preparation of the template–monomer complex, easy removal of the templates from the polymers, fast binding of templates to MIPs and its potential application to a wide range of target molecules. However, it is not free of some drawbacks because the template–monomer interactions are governed by an equilibrium process. Thus, to displace the equilibrium toward the formation of the template–monomer complex, a high amount of monomer is always used. Consequently, the excess of free monomers is randomly incorporated to the polymeric matrix, leading to the formation of nonselective binding sites.

MISPME

Polymerization strategies of MISPME

The selectivity required in SPME can be provided through molecular imprinting, as was first demonstrated by Koster *et al.* (48), who prepared a fiber coated with a clenbuterol-imprinted polymer. Since then, more polymerization strategies of MISPME have been developed in different formats. They can primarily be classified as MIP-coated fibers (polymeric membranes) and MIP rodlike fibers (polymeric monoliths). A schematic representation of MIP-coated fibers of MISPME is shown in Figure 3.

MIP-coated fibers

This strategy is based on a thin fused silica fiber coated with an MIP, and the MIP-coated fiber is used as an SPME fiber to perform in a sample pretreatment procedure. In 2001, Koster *et al.* (48) first reported the preparation of MIP-coated silica fibers for the SPME of brombuterol from human urine. In this

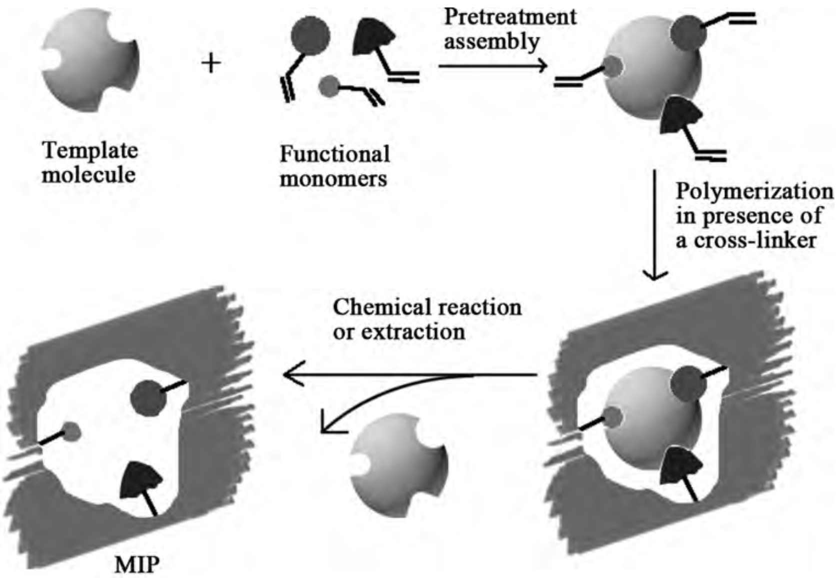

Figure 2. Diagram showing the synthesis of MIPs.

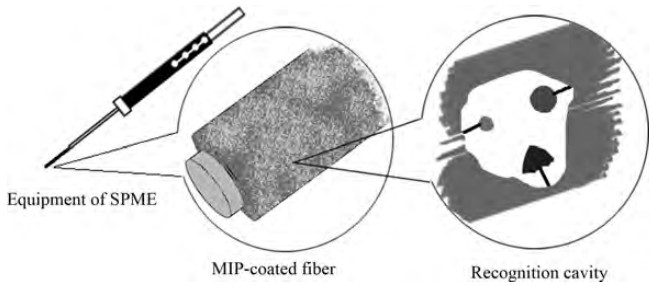

Figure 3. Schematic representation of MIP-coated fiber of MISPME.

work, fibers coated with a 75-μm layer of methacrylate polymer imprinted with clenbuterol were obtained in a reproducible manner and successfully used for the selective extraction of brombuterol from urine samples. Hu and co-workers (49) developed novel prometryn MIP-coated SPME fibers with good selectivity for the extraction of triazines. The fibers, coupled with high-performance liquid chromatography (HPLC), were applied to the determination of prometryn and its analogues with similar structures such as atrazine, simetryn, terbutylazin, ametryn, propazine and terbutryn in the spiked soy bean, corm, lettuce and soil samples. Subsequently, Hu *et al.* prepared tetracycline-imprinted coating fibers using a similar method (50). The fibers are suitable for the simultaneous multi-residue analysis of trace oxytetracycline, tetracycline, doxycycline and chlortetracycline in complicated samples. Tan *et al.* (51) developed a simple preparation approach for bisphenol A (BPA) MIP-coating with controlled thickness on fused-silica capillaries. The MIP-coated fibers were successfully used for the selective extraction of BPA spiked into tap water, human urine and liquid milk samples, and satisfactory recoveries were obtained. Unfortunately, the analytical performance using these fibers is not satisfactory for applicability to real environmental matrices at its present stage.

Several novel polymerization methods can be used to prepare MIP-coated fibers of MISPME. An improved multiple co-polymerization technique was first developed to prepare MIP-coated fibers with propranolol and 17β-estradiol as templates by Hu and co-workers (52, 53). The obtained fibers exhibited excellent characteristics such as high porosity and good thermal and chemical stability. The surface reversible addition-fragmentation chain transfer (RAFT) polymerization method was first applied by Hu *et al.* (54) to the preparation of MIP-coated SPME fibers with Sudan I as a template. In surface RAFT polymerization, the radical growing and chain propagation are controllable, so an ultra-thin MIP coating could be obtained with a homogeneous structure and controlled composition. The scanning electron micrographs (SEMs) of a series of MIP-coated fibers are shown in Figure 4.

MIP monolithic fibers

In this strategy, a completely different and much simpler approach is demonstrated for the preparation of imprinted fibers. It is based on the direct synthesis of molecularly imprinted polymeric fibers (monoliths) using capillaries as molds, which are etched away after polymerization. In 2007, this approach for the preparation of imprinted fibers was proposed by Turiel and co-workers (55). The performance of the new imprinted fibers was assessed for the SPME of triazines from environmental and food samples. In this regard, Djozan and Ebrahimi (56) have reported detailed studies on the thermal stability of monolithic fibers. Based on their results, these monolithic fibers are stable up to 280°C (Figure 5). He *et al.* (57) reported phthalate MIP monolithic fibers of SPME, which could be coupled directly to GC and GC–mass spectrometry (MS) for the extraction of trace phthalates in complicated samples.

Development of MISPME

However, despite the rapid development of this methodology, both types of fibers are fragile, lacking porosity and thereby provide poor accessibility of the target analyte to binding sites.

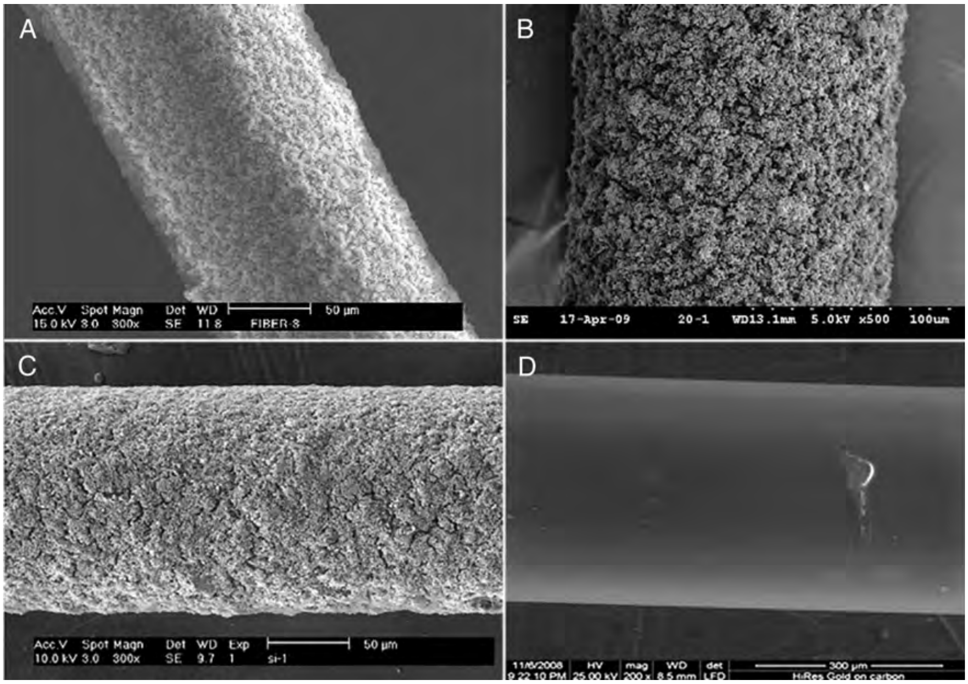

Figure 4. SEMs of MIP-coated fibers prepared from: Hu *et al.* (49) (A); Hu *et al.* (53) (B); Hu *et al.* (50) (C); Tan *et al.* (51) (D).

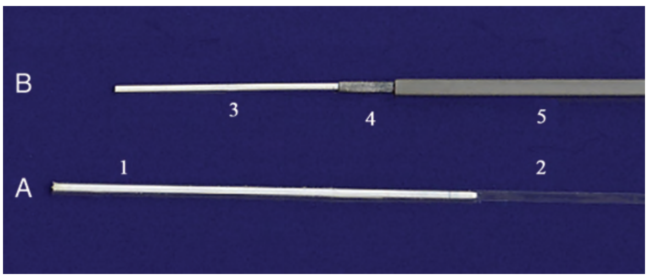

Figure 5. Prepared MIP monolithic fiber: before mechanical withdrawals from glass mold (A); after mechanical withdrawals from glass mold (B); fiber in mold (1), glass mold (2), fiber in homemade SPME syringe (3), inner needle of syringe (4) and outer needle (5). Reprinted from Djozan and Ebrahimi (56).

Therefore, stability and low capacity of such fibers need major improvements. In 2008, Prasad *et al.* (58) prepared ascorbic acid (AA) MIP-coated SPME fibers using silylated polymethyl methacrylic acid (PMMA) fibers as material supported by sol-gel technology. In this case, the organic–inorganic–organic interpenetrating network caused by the sol-gel adhesion between the MIP film layer and fiber surface were expected to impart a high level of porosity and enhanced surface area of SPME fiber. The fiber was used in combination with a complementary MIP sensor for a better signal amplification to attain a stringent detection limit of in highly diluted aqueous, biological and pharmaceutical samples, without any cross-reactivity or matrix effect. Subsequently, they prepared dopamine MIP-brush coating using a similar method. Li and co-workers (59) extended the potential of the conventional sol-gel technique for the fabrication of an organically modified silicate (ORMOSIL) SPME stationary phase molecularly imprinted with decabromodiphenyl ether from phenyltrimethoxysilane and tetraethoxysilane. The imprinted ORMOSIL-SPME device was successfully applied to monitor polybrominated diphenyl ethers (PBDEs) in municipal wastewaters.

The drawback of the fragile and short lifetime of the fused silica fiber used as the support material restricts the development of MISPME. Therefore, the use of metal wires as fiber supports with high mechanical stability makes this technique more robust for routine analysis. Djozana *et al.* (60) first reported the fabrication of an SPME fiber obtained through photo-polymerization [under ultraviolet (UV) irradiation] of a pre-polymer solution on the surface of silylated anodized aluminum wire, to achieve a chemically bonded coating with ametryn-imprinted polymer. This fiber was evaluated and subsequently used for the extraction of ametryn and its structural analogs for GC and GC–MS analyses.

To improve the selectivity and extraction efficiency of MISPME fibers, a novel electrochemically controlled MISPME was developed, which used conductive polymers such as molecularly imprinted PPy and its derivatives as the extraction phase. Ameli and Alizadeh. (61) reported the electrodes, which were electrochemically coated with overoxidized polypyrrole (OPPy) films of salicylate-MIP. The OPPy film as the SPME absorbent was applied for the selective cleanup and quantification of trace amounts of salicylate from physiological samples. Liu *et al.* (62) developed an electrochemically enhanced SPME (EE-SPME) approach based on a composite coating of molecularly imprinted PPy (MIPPy)/MWCNTs on platinum wire for the selective extraction of fluoroquinolone antibiotics (FQs) in aqueous samples. This method showed good selectivity and higher extraction efficiency of the analytes than traditional SPME.

The application of an MIP-coated fiber was disturbed by the water-compatibility problem as a result of the competition

Table I
Application of MISPE to Real Samples

Analyte	Samples	Sample preparation method	Analytical system	Analytical parameters	Reference
Brombuterol	Human urine	MIP-coated fiber SPME	LC–ECD	Limit of detection (LOD): $\sim$10 ng/mL	48
Propranolol	Serum	In-tube MISPME	HPLC–UV	Relative standard deviation (RSD): 7.6% LOD: 0.32 μg/mL	67
Triazines	Soybean, corn, lettuce and soil	MIP-coated fiber SPME	HPLC–UV	Recoveries: 75.5–113.2% RSD: 2.9% LODs: 0.012–0.090 μg/L	49
Triazines	Soil, potato and pea	MIP monolithic fiber SPME	HPLC–UV		55
Diacetylmorphine and analogous compounds	Street heroin	MIP monolithic fiber SPME	GC–flame ionization detection (FID), GC–MS	RSDs: 5.5–15.6% LODs: 1–300 ng/mL	68
AA	Human blood serum and pharmaceutical samples	MIP-coated fiber SPME	Polarographic analyzer	RSD: 2.3% LOD: 0.0396 ng/mL	58
Tetracyclines	Chicken feed, chicken muscle and milk	MIP-coated fiber SPME	HPLC equipped with a fluorimetric detector	Recoveries: 71.6–93.7% RSD: 6.7% LODs: 1.0–2.3 μg/L	50
Triazines	Water, onion and rice	MIP monolithic fiber SPME	GC–FID, GC–MS	Recoveries: >90% RSDs: <10% LODs: 20–90 ng/mL	56
PBDEs	Municipal wastewaters	MIP-coated fiber SPME	HPLC–μ electron capture detector, GC–μECD and GC–MS	RSD: 3.7–17.9% LODs: 0.2–8.8 pg/mL	59
BPA	Tap water, human urine and liquid milk	MIP-coated fiber SPME	HPLC equipped with a diode array detector	Recoveries: 81.6–92.5% RSDs: 4.7–9.9% LODs: 1.7–38.9 ng/mL	51
β-Blockers	Urine and plasma	MIP-coated fiber SPME	HPLC–UV	Recoveries: 75.2–88.2% and 76.4–88.8% RSD: 6.4% LODs: 3.8 and 6.9 μg/L	52
Dopamine	Human blood serum and cerebrospinal fluid	MIP-coated fiber SPME	Voltammetric analyzer	Recoveries: 99.7–101.3% RSD: 0.6% LOD: 0.018 ng /mL	66
Estrogens	Fish and shrimp	MIP-coated fiber SPME	HPLC–UV	Recoveries: 80.0–94.1 % RSDs: 3.8–8.3% LODs: 0.98–2.39 μg/L	53
8-Hydroxy-2′-deoxyguanosine	Human urine	In-tube MISPME	Capillary electrophoresis (CE)	Recoveries: 85% ± 3.5% RSD: 3.7% LOD: 2.6 nmol/L	69
2,2′-Dipyridine	Tap water, laboratory wastewater and human urine	Metal complex imprinted polymer (CIP)-coated fiber SPME	HPLC–UV	Recoveries: 80.3–103.3% RSDs: 5.5–8.9% LOD: 2.0 μg/L	70
Triazines	Water, rice, maize and onion	MIP-coated fiber SPME	GC	Recoveries: >85% RSDs: <10% LODs: 9–85 ng/mL	60
Anabolic steroids	Human urine	MIP-coated fiber SPME	GC–MS	Recoveries: 80.1–108.4% RSDs: 4.2–14.7% LODs: 0.02–0.1 ng/mL	71
Phthalates	Environmental water	MIP monolithic fiber SPME	GC–MS	Recoveries: 94.54–105.34% RSDs: 1.50–8.04% LODs: 2.17–20.84 ng/L	57
Salicylate	Urine and serum	MIP-coated fiber SPME	Fluorescence spectrometer	Recoveries: 96–108% RSDs: 2.3–3.8% LOD: 4×10^{-8} mol/L	61
Metolachlor, propisochlor and butachlor	Real river water and farm water	MIP-coated fiber SPME	HPLC–UV	Recoveries: 75.6–95.4 and 79.3–97.9% RSDs: 3.1, 4.3 and 4.5% LODs: 0.2, 1.4 and 2.8 μg/L	63
Thiabendazole	Orange juice	SLM-protected MIP-coated fiber SPME	HPLC equipped with a fluorimetric detector	Recoveries: 6.9–7.0% RSDs: 6.6–9.7% LOD: 4 μg/L	65

(*continued*)

Table I Continued

Analyte	Samples	Sample preparation method	Analytical system	Analytical parameters	Reference
Acetaldehyde	Mineral water, coke, orange juice and pineapple juice	MIP monolithic fiber SPME	GC–MS	Recoveries: 75.8–90.0% RSDs: 8–15% LOD: 0.01 μg/mL	72
FQs	Urine and soil	EE MIP-coated fiber SPME	HPLC	Recoveries: 85.1–95.5% RSDs: 2.6–6.5% LODs: 0.5–1.9 μg/L	62
Chlorogenic acid	Medicinal plant	MIP-HF-SPME	HPLC–UV	Recoveries: 84.8–97.2% RSD: 0.38% LOD: 0.08 ng/mL	73
Estrogens	Urine, milk and skin toner	MIP-coated fiber SPME	HPLC	Recoveries: 81.9–99.8 % RSDs: 4.1–7.9% LODs: 0.08–0.25 μg/L	64
Sudan dyes	Chili tomato sauce and chili pepper	MIP-coated fiber SPME	HPLC–UV and LC–tandem MS	Recoveries: 88.4–96.1% RSDs: 6.3–11.7% LODs: 0.21–0.75 ng/g	54
Sulfamethazine	Milk	MIP-DSPME	CE–UV	Recoveries: 89–110% RSD: 4.2% LOD: 1.1 μg/L	74
Catecholamines	Urine	MIP monolithic fiber SPME	CE–UV	Recoveries: 85–1.3% RSDs: 5.9–9.8% LODs: 4.8–7.4 nmol/L	75
Ephedrine and pseudoephedrine	Serum and urine	MIP monolithic fiber SPME	CE–UV	Recoveries: 91–104% RSDs: 3.8–9.1% LODs: 0.00096 and 0.0011 μg/mL	76
2,4,6-Trinitrotoluene	Rubble samples taken from post-blast scenarios	MIP-coated fiber SPME	GC–MS	RSDs: <10% LOD: 60 ng/kg	77

between analytes and water for the recognition cavities in MIP. A novel liquid–liquid–solid microextraction (LLSME) device was designed by Hu and co-workers (63) to solve the water-compatibility problem of the MIP-coated SPME fiber. In this method, the aqueous sample was first added into the lower room of the LLSME device, and then the rubber stopper was utilized to adjust the liquid level of the aqueous solution to the boundary between the upper and lower rooms. After that, a small amount of organic solvent was loaded into the upper room, and the two-phase system was stirred to reach liquid–liquid microextraction equilibrium. Subsequently, an MIP-coated fiber was immersed into the organic solvent for SPME under stirring. The practical applicability of this method was validated with samples of spiked real river water and farm water. The results demonstrated the high enrichment capabilities and selectivity for trace metolachlor, propisochlor and butachlor in aqueous samples. A novel sample preparation technique termed dynamic liquid–liquid–solid microextraction (DLLSME) was developed and online coupled to HPLC for the direct extraction, desorption and analysis of trace estrogens in complex samples by Zhong *et al.* (64). DLLSME consists of the aqueous donor phase, the organic medium phase and the molecularly imprinted polymer filaments (MIPFs) as the solid acceptor phase. The organic solvent with lesser density was directly added on top of the aqueous sample, and the dynamic extraction was performed by circulating the organic solvent through the MIPFs inserted into a polyetheretherketone (PEEK) tube, which served as an extraction and desorption chamber. After this, the extracted analytes on the MIPFs were online desorbed and then introduced into the HPLC for analysis. Barahona *et al.* (65) reported an organic supported liquid membrane (SLM)-protected MISPME procedure to solve the inherent limitations of MIPs, such as target recognition in aqueous media. In this work, MIP-coated fibers were located inside a polypropylene hollow capillary and protected by an organic solvent immobilized as a thin SLM in the pores of the capillary wall. The extraction procedure involved two simultaneous processes: liquid phase microextraction (LPME) using polypropylene hollow fibers (HF-LPME) of the analytes from the sample to an organic acceptor solution through an SLM; and SPME of the analytes from the organic acceptor solution to a MIP-coated fiber inside the polypropylene capillary.

Applications of MISPME

Table I lists the primary applications of MISPME to the extraction of analytes in real samples, published before 2012. As shown in Table I, most of these studies were devoted to the analysis of pharmaceuticals in biological samples, such as ascorbic acid (58) and dopamine (66) in human serum and urine. Figure 6 shows the chromatograms of brombuterol in human urine and a spiked sample subjected to MISPME with liquid chromatography–electron capture detector (LC–ECD) (48). Furthermore, applications of MISPME to the extraction and determination of pesticide residues have drawn much attention, due to their wide distribution in environmental samples and their pronounced toxicity and effects upon aquatic organisms. Several studies have dealt with the extraction and determination of triazines (49, 56, 60), PBDEs (59) and thiabendazole (65).

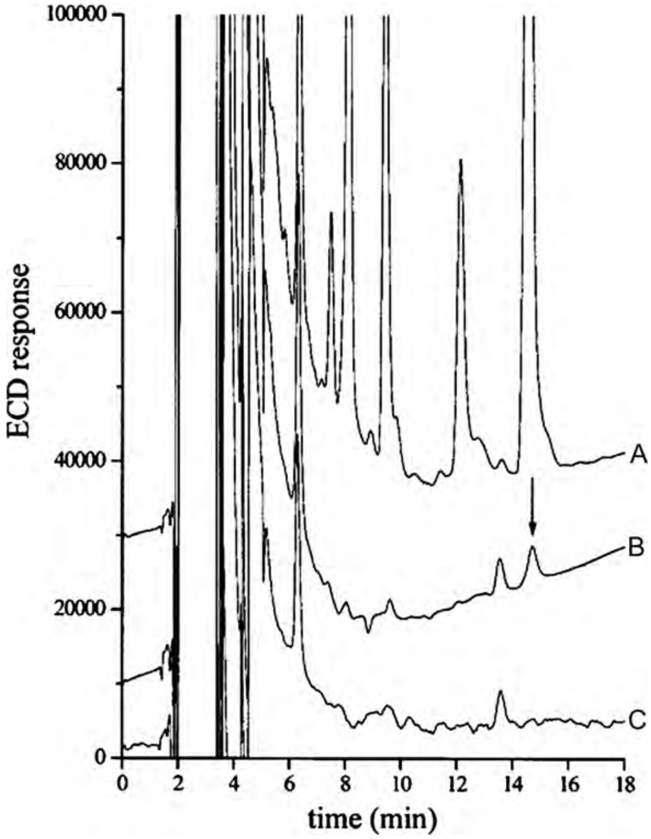

Figure 6. LC–ECD chromatograms of the extraction (45 min) of human urine with MIP-coated fiber: blank urine (A); spiked urine (100 ng/mL brombuterol) and washing of the fiber (B); blank urine (C). Fibers were washed in 200 μL of acetonitrile for 5 min. The brombuterol peak is indicated by the arrow. Reprinted from Koster *et al.* (48).

Outlook and future trends

Because of its advantages, including high selectivity, stability and reusability, MISPME is successfully used as a novel microextracton technique for selectively enriching analytes from different real samples, such as environmental, biological, food and other samples. However, some features still need improvement. Future developments of MISPME are primarily represented as two aspects: (i) new MISPME coating materials with higher selectivity and responses to a broad range of analytes will be developed to expand the application range of this technique; (ii) the performance of established MISPME techniques will be improved, which will ease the implementation of MISPME in analytical laboratories. To conclude, the development of new or improved MISPME formats and coatings may still be the primary topics in this field in the future.

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